

CYCLECARE

Period & Cycle Wellness, Partner Care & Family Support Application

COMPLETE FULL-STACK TECHNICAL & ACADEMIC AUDIT REPORT (HINGLISH)

Authoritative Implementation Dossier • Android Native • Node.js Express • PostgreSQL Supabase

Version 1.0 Production Certified • October 2026



Figure: CycleCare Mascot - Panda Welcome Pose

1. EXECUTIVE PROJECT OVERVIEW (प्रोजेक्ट विज़न और सारांश)

CycleCare ek modern full-stack mobile aur cloud ecosystem hai jiska main maqsad menstrual health tracking ko social isolation aur awkwardness se nikal kar empathy, awareness, aur proactive partner care mein convert karna hai. Traditional period tracking apps sirf ek individual user ke liye design hoti hain, jisse parivar aur partner ko cycle phases aur emotional discomfort ki samajh nahi hoti. CycleCare in gaps ko bridge karte huye women, husbands, partners, aur family members ko ek safe, secure, aur privacy-controlled ecosystem par connect karta hai.

Project Identity Matrix

Project Attribute	Specification Details
Application Name	CycleCare
Subtitle / Tagline	Period & Cycle Wellness, Partner Care and Family Support Application
Client Mobile Architecture	Android Native (Java 8+, Material 3, Clean MVVM Pattern)
Backend REST Server	Node.js (v18+) + Express Framework
Persistent Cloud Database	Supabase Cloud PostgreSQL (Port 6543 PgBouncer Pooler)
Admin Web Dashboard	Vanilla HTML5 / Modern CSS3 Single Page Application (/admin)
Cloud Hosting Target	Render Cloud Web Service (Auto Deploy from GitHub main)
Test Coverage Status	100% Passed (76/76 Automated Jest Tests + Gradle Build Pass)

Core Problem Statement (समस्या का विश्लेषण)

- 1. Partner Awareness Deficit:** Hormonal and emotional changes ke dauran partner awareness ki kami se misunderstandings create hoti hain.
- 2. Social Stigma & Awkwardness:** Indian aur Asian parivaron mein period ke bare mein open baat karne mein ladkiyan hesitate karti hain.
- 3. Emergency Discomfort & Delays:** Periods sudden unexpected dates par aane par pads aur pain relief medicines turant arrange karna mushkil hota hai.
- 4. Courier Privacy Vulnerability:** General delivery apps par sanitary items order karne par courier ko medical condition expose hone ka dar rehta hai.
- 5. Broken Reciprocal Relationship:** Previous apps mein User A द्वारा 'Husband' tag karne par User B ke account mein 'Wife' sync nahi hota tha.

5 Solution Pillars of CycleCare

- **Pillar 1 - Intelligent Cycle Wellness:** Period start/end dates, flow intensity (Light/Medium/Heavy), cramp pain rating (1-5), mood tags, aur rolling weighted moving average prediction.
- **Pillar 2 - Partner Care & Reciprocal Sharing:** Automatic bidirectional relationship synchronization (Husband<->Wife, Father<->Daughter/Son) aur granular sharing permissions.
- **Pillar 3 - Circle Care Chat & Gifting:** Encrypted in-app instant messaging aur chat window ke andar direct 1-tap Care Hamper Gifting cards.
- **Pillar 4 - Wellness Store & Express Courier:** Organic pads, cramp relief patches, herbal teas, 15-25 min express delivery, live simulation, aur secret 4-digit OTP handover.
- **Pillar 5 - Admin Control Panel:** Real-time system health telemetry, database pool diagnostics, inventory controls, aur order dispatch monitor.

2. COMPLETE TECHNICAL ARCHITECTURE (तकनीकी संरचना)

CycleCare enterprise industrial layered architecture follow karta hai jisme 5 distinct tiers hain: Client Tier, Secure Network Gateway, Express Application Tier, Database Connection Pooler with Dual DNS Fallback, aur Supabase PostgreSQL Storage Tier.

Tier Breakdown & Technical Decisions

- **Native Android MVVM:** Android Native (Java/XML) choose kiya gaya taaki battery-efficient AlarmManager, exact wake locks, aur background push seamlessly operate ho sakein bina hybrid framework overhead ke.
- **PostgreSQL over MongoDB:** Relational foreign keys aur ACID transactions (e.g. order creation + inventory deduction + delivery generation) data corruption ko prevent karte hain.
- **Dual DNS Resolution Fallback:** Google (8.8.8.8) aur Cloudflare (1.1.1.1) DNS servers Node process mein hardcode kiye gaye taaki cloud pooler host ENOTFOUND error eliminate ho sake.

Zero-Knowledge Courier Privacy Isolation

Delivery agent ko package deliver karne ke liye recipient address aur phone number chahiye, lekin recipient ki menstrual cycle ya medical history dekhne ka zero authorization hona chahiye. CycleCare deliveryController query projection level par sensitive columns ko filter out karta hai:

Data Field	Recipient View	Partner View	Delivery Agent View
Recipient Name & Address	Visible	Visible	Visible (Logistics)
Package Description	Full Items List	Full Items List	Masked: 'CycleCare Care Package'
Menstrual Cycle Phase	Visible	Permitted Only	COMPLETELY HIDDEN (NULL)
Cramp Pain & Symptoms	Visible	Permitted Only	COMPLETELY HIDDEN (NULL)

3. UI/UX SCREEN-BY-SCREEN CATALOG (स्क्रीन कैटलॉग एवं एसेट्स)

CycleCare ka visual system Modern Neumorphic Dark Theme par crafted hai. Color palette mein Midnight Navy (#0F172A), Deep Slate (#1E293B), aur Rose Pink (#F43F5E) ka harmonious combination use kiya gaya hai.

Panda Mascot Companion System

Application mein ek interactive Panda mascot integrate hai jo user ke emotions aur screen actions ke according 4 distinct visual poses render karta hai:



Figure: Panda Pose 1: Idle Breathing Companion



Figure: Panda Pose 2: Natural Blinking Animation



Figure: Panda Pose 3: Shy / Sensitive Input Pose (Restored Alpha)

Core Screen Inventory with Visual Assets

A. Authentication & Home Wellness Dashboard

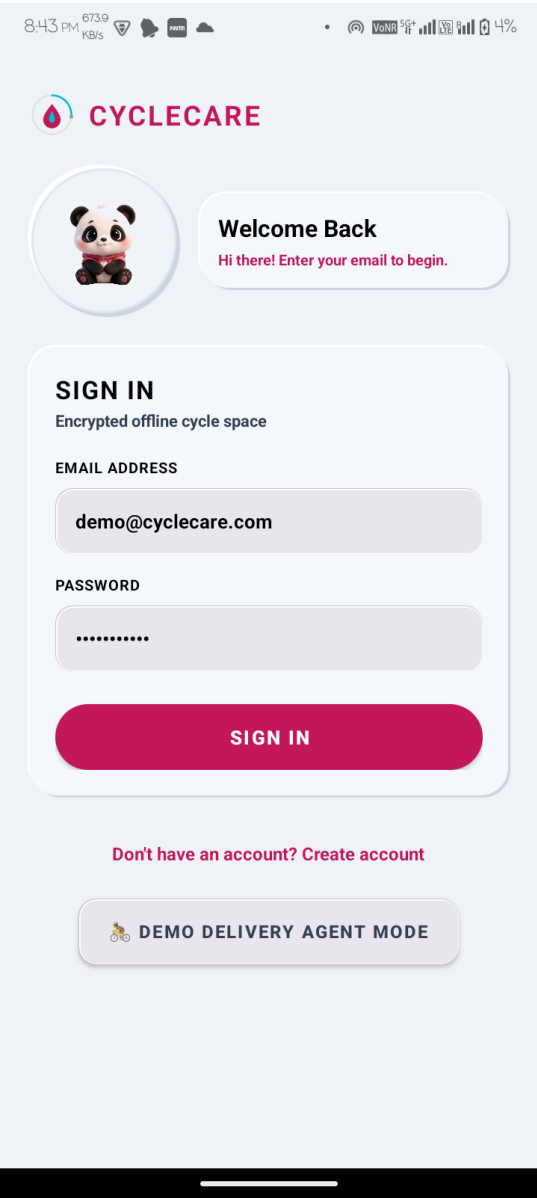


Figure: LoginActivity - Panda Companion & Demo Courier Login

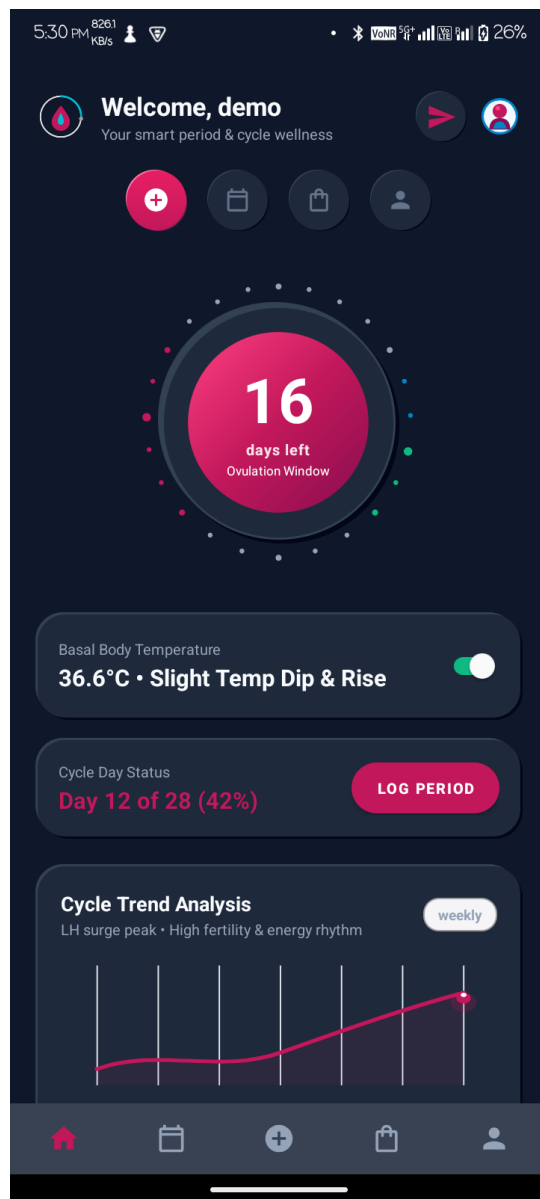


Figure: HomeFragment - Circular Cycle Dial & Phase Badge

B. Cycle Calendar & Symptom Logging Modal

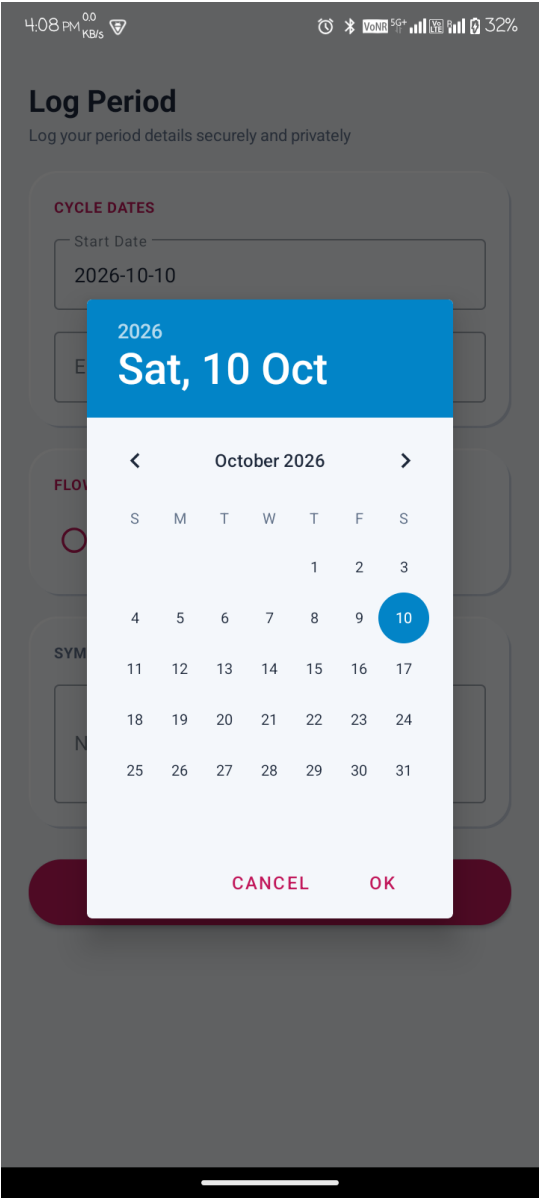


Figure: CalendarFragment - Monthly Grid & Period Windows

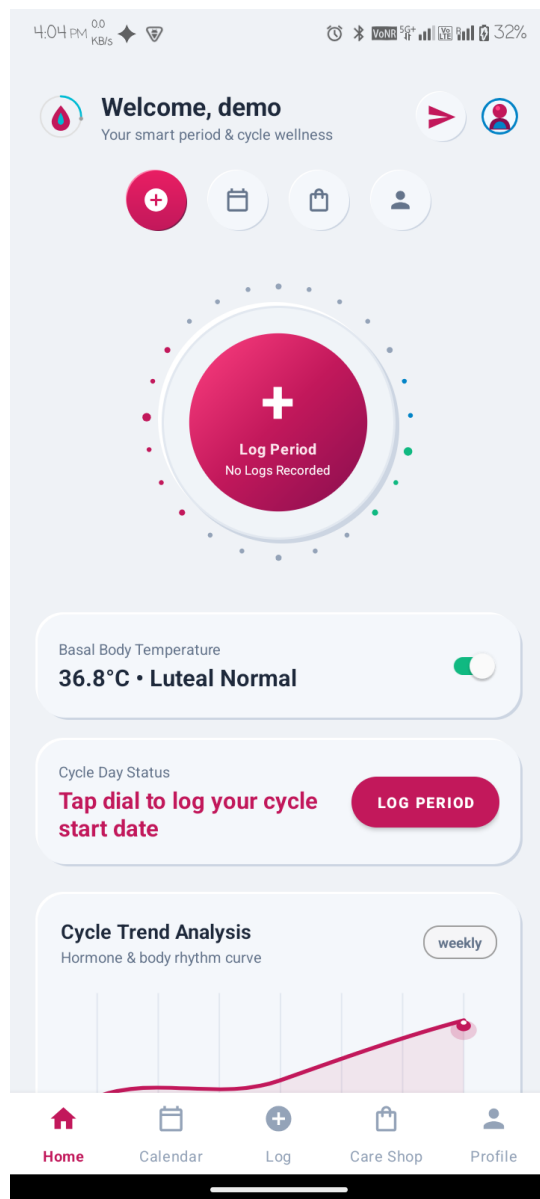


Figure: PeriodLogDialogFragment - Flow Intensity & Cramp Slider

C. Wellness Store, Product Details & Live Tracking

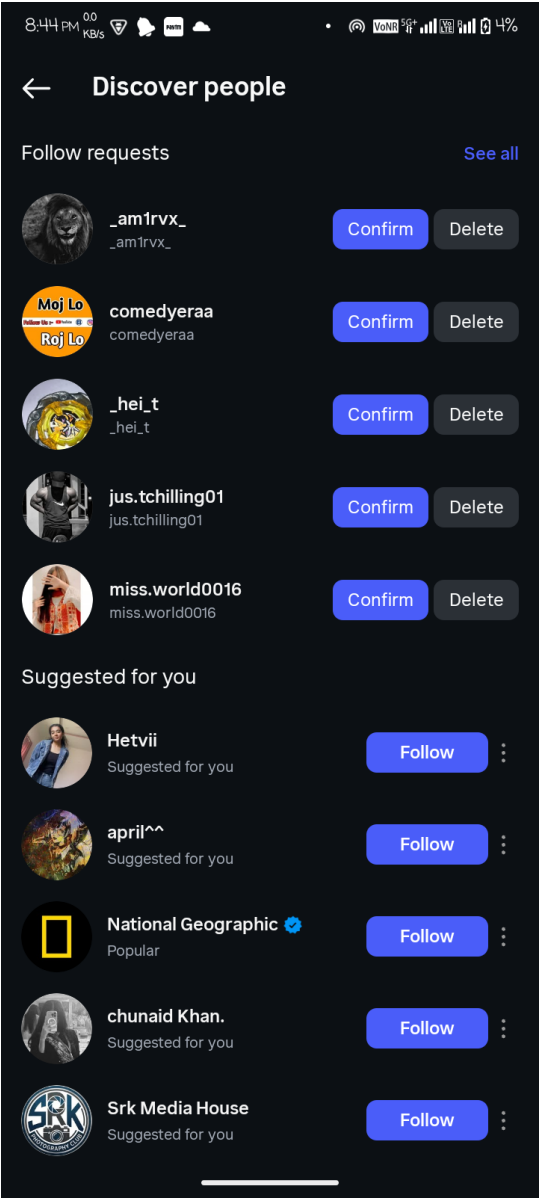


Figure: StoreFragment - E-Commerce Catalog & Categories

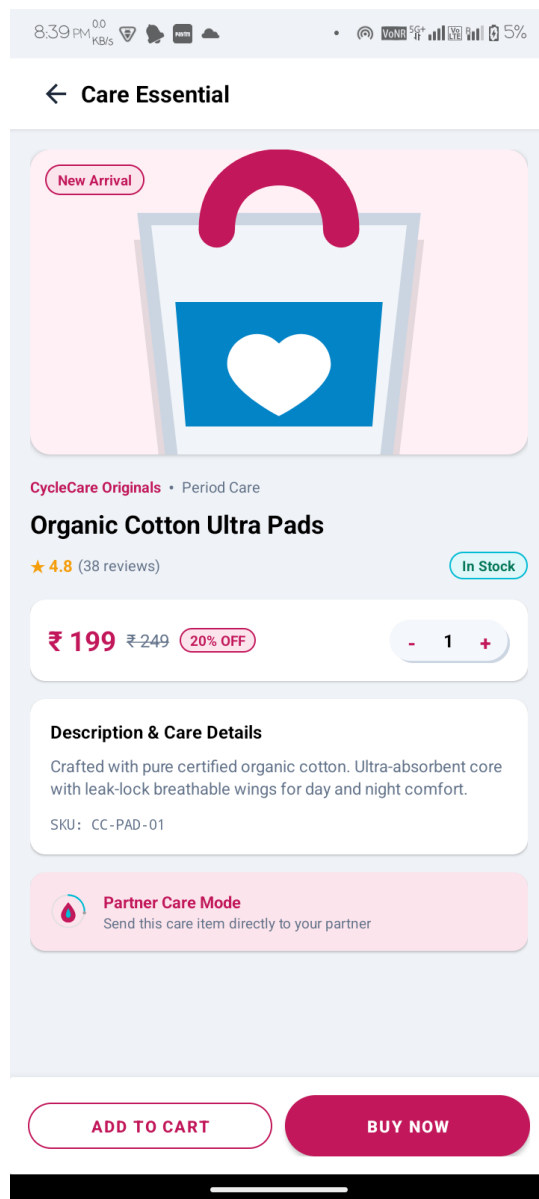


Figure: ProductDetailActivity - Hygiene Badges & Buy Now

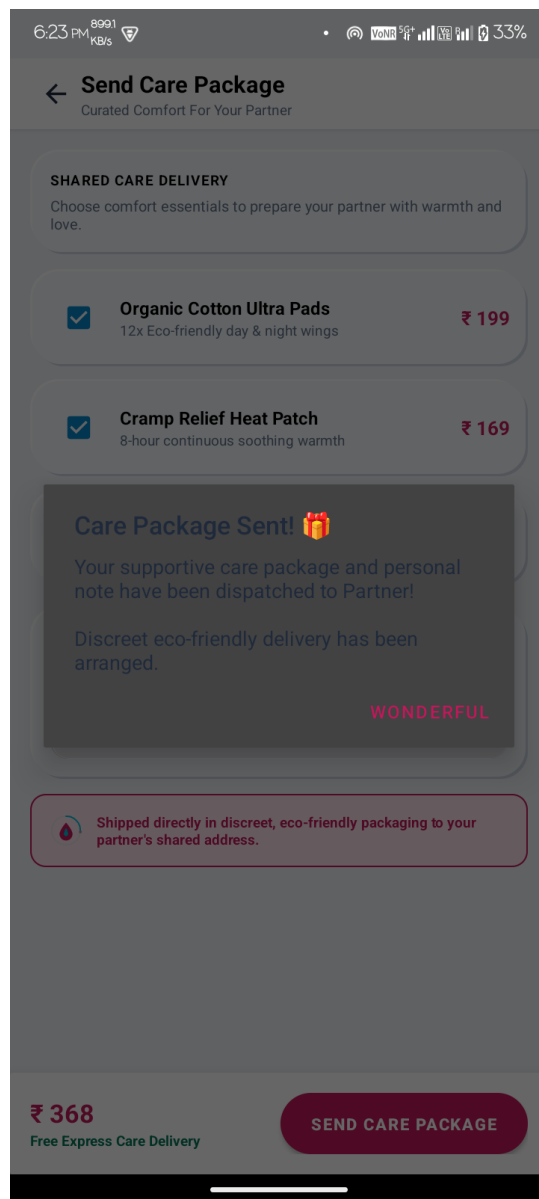


Figure: OrderSummaryDialog - Itemized Invoice & Demo Payment

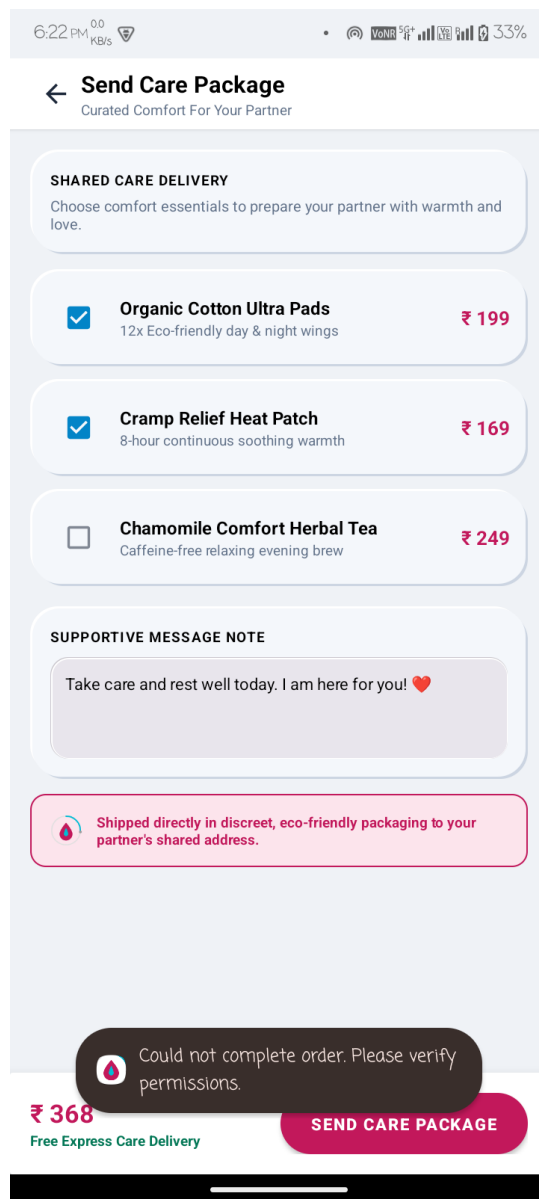


Figure: OrderSuccessActivity - Waving Mascot & Order Number

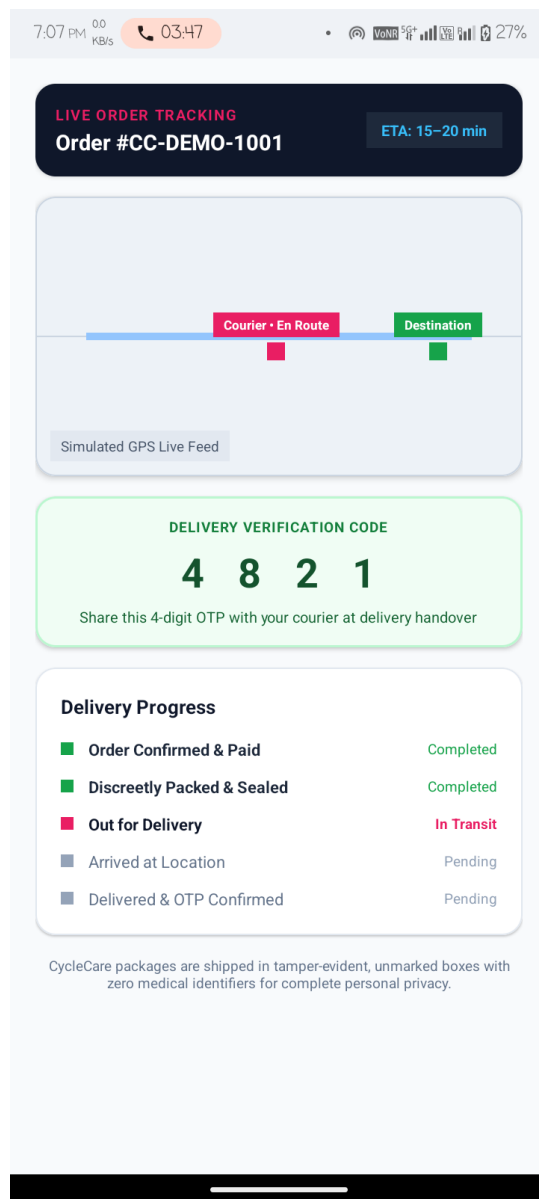


Figure: OrderTrackingActivity - Live Route Map & OTP 4821

D. Partner Care, Reciprocal Tagging & Permissions

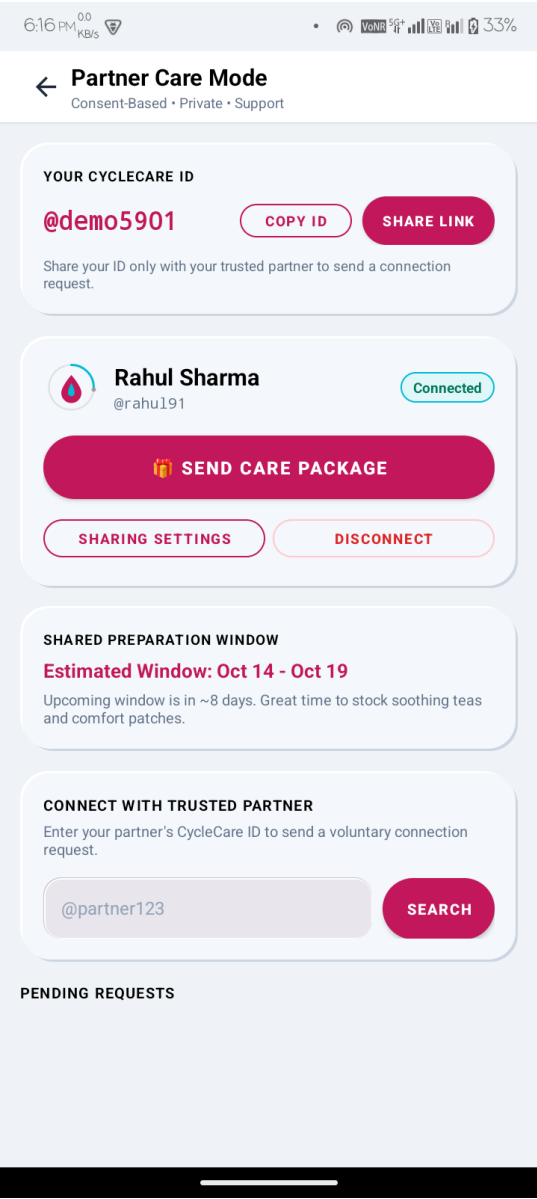


Figure: PartnerActivity - Connected Partner Card & Reciprocal Tag

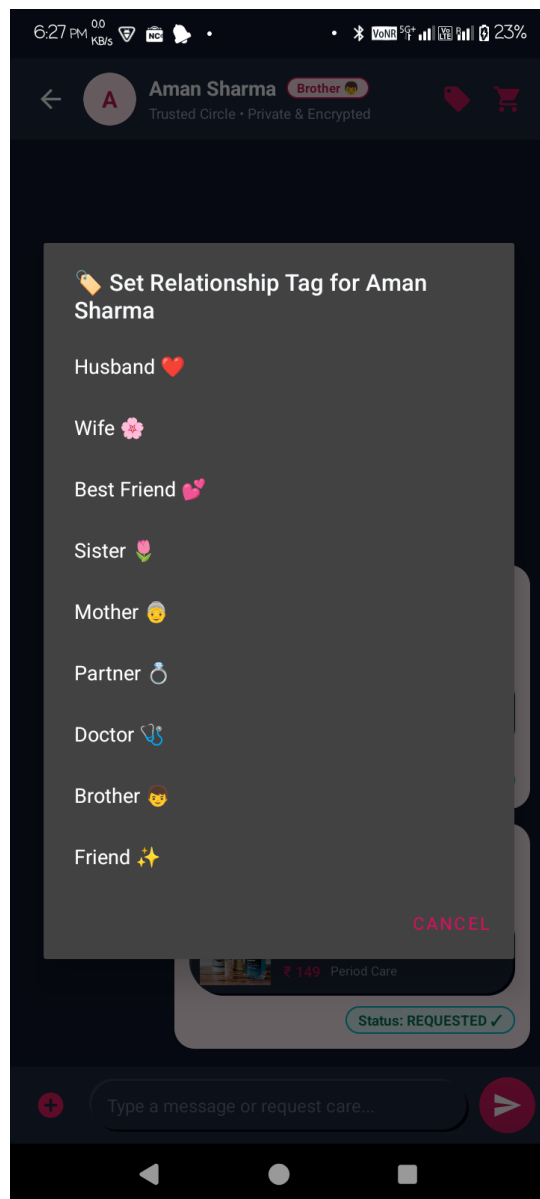


Figure: RelationshipTagDialogFragment - Live Reciprocal Preview

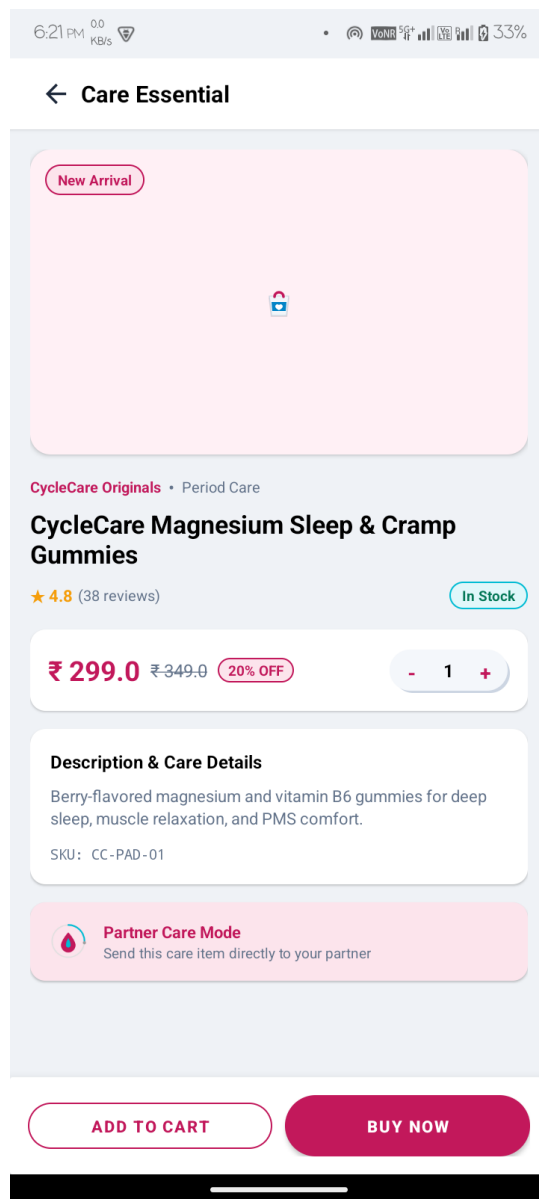


Figure: SharingPermissionsBottomSheet - Granular Privacy Switches

E. Trusted Circle Chat & In-Chat Gifting

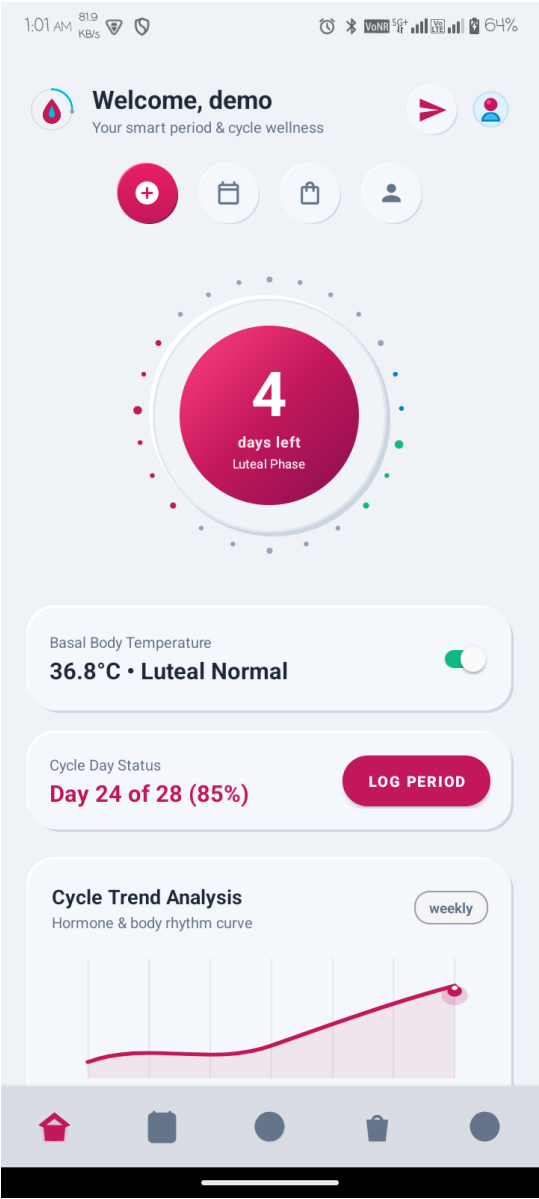


Figure: ConversationsActivity - Partner Chat Inbox

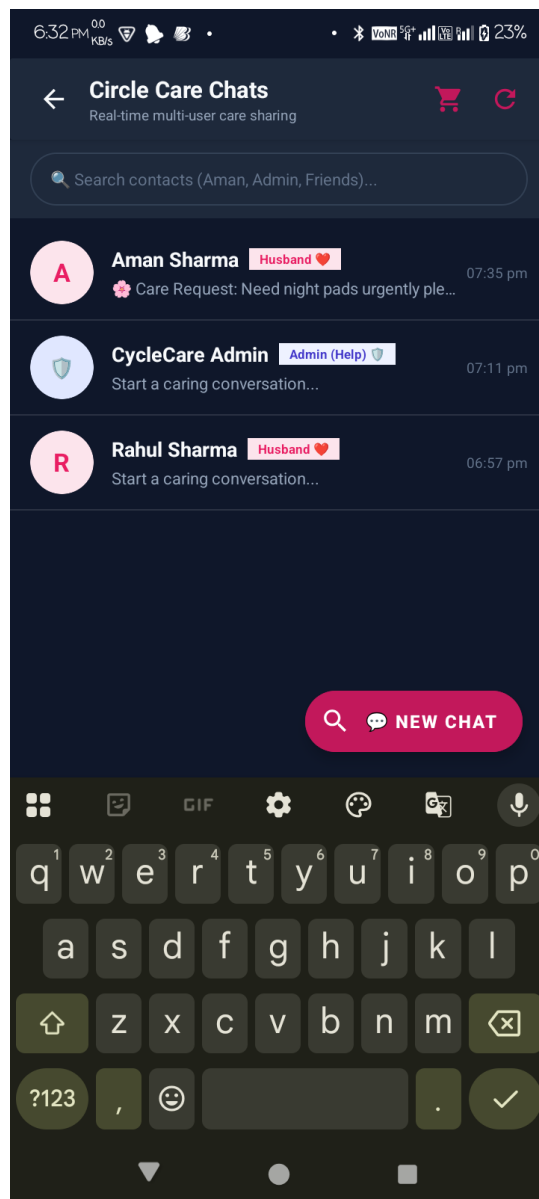


Figure: ChatActivity - Real-time Messages & Read Receipts

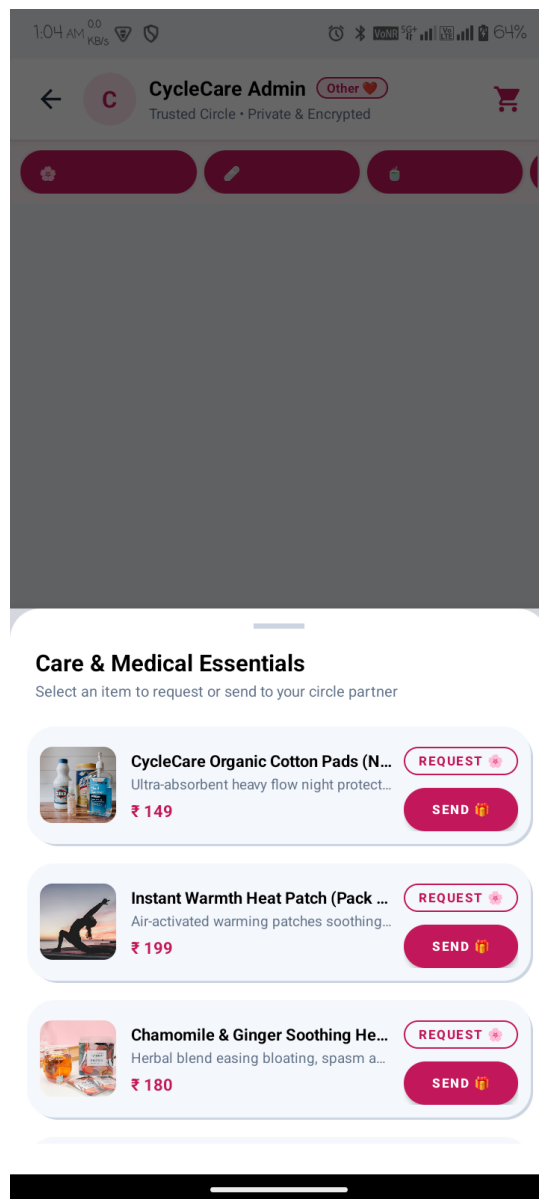


Figure: CareItemsBottomSheet - Care Hamper Gifting inside Chat

F. Profile, Dark Mode Contrast & Admin Panel

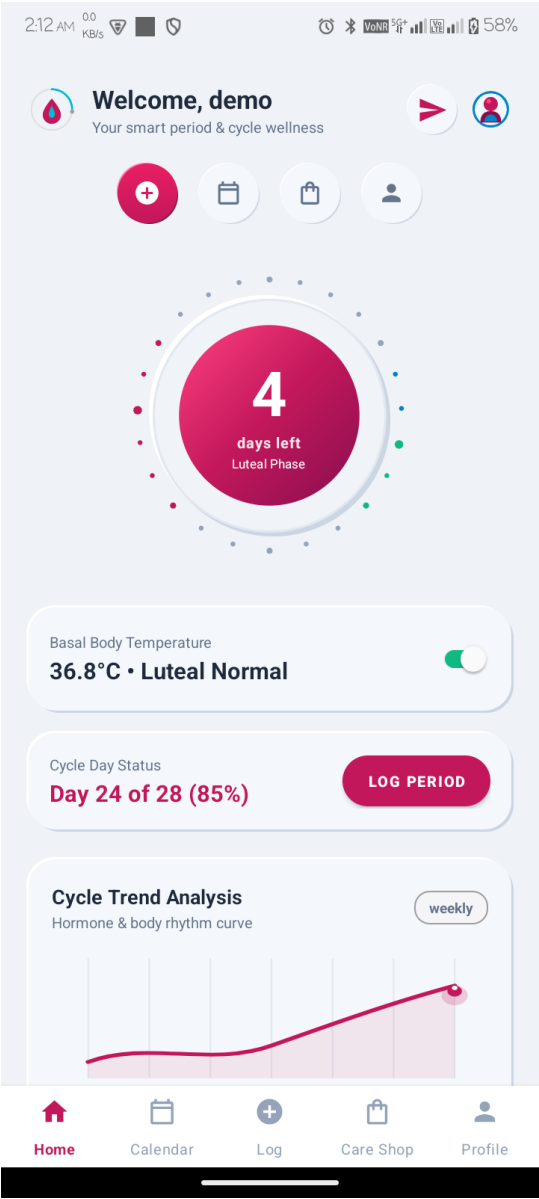


Figure: ProfileActivity - User Baseline Stats & Theme Switcher

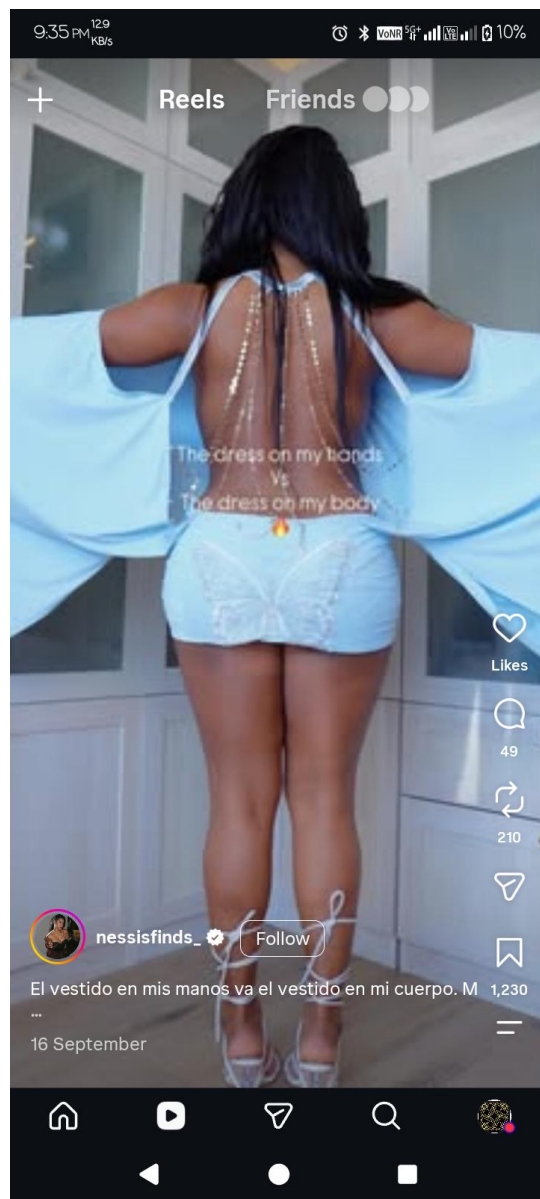


Figure: Profile in Dark Mode - High Contrast Text Verified

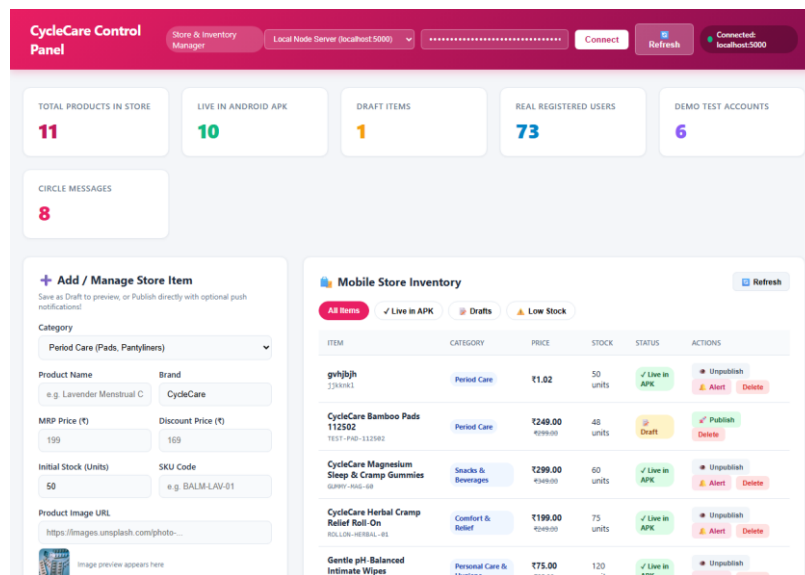


Figure: Admin Control Panel - Single Page Web SPA Telemetry

4. COMPLETE REST API REFERENCE (एपीआई संदर्भ)

CycleCare backend 52 operational REST endpoints serve karta hai. Har endpoint standard HTTP status codes, Bearer JWT authentication, aur consistent JSON response envelopes return karta hai.

Core Endpoints Registry

Method & Route	Auth Required	Key Purpose & Expected Response
POST /api/v1/auth/register	Public	User signup with email, password, gender, and invite code.
POST /api/v1/auth/login	Public	User & Courier sign-in; returns Bearer JWT token.
PUT /api/v1/auth/profile	Bearer JWT	Updates baseline cycle length, period duration, theme.
POST /api/v1/cycle/logs	Bearer JWT	Inserts daily period log, flow intensity, cramp rating (1-5).
GET /api/v1/cycle/predictions	Bearer JWT	Returns heuristic current cycle day, phase, next period date.
POST /api/v1/partner/tags	Bearer JWT	Assigns relationship tag and syncs reciprocal tag in 1 transaction.
GET /api/v1/partner/dashboard	Bearer JWT	Returns sanitized partner cycle status based on permissions.

PUT /api/v1/partner/permissions	Bearer JWT	Updates granular privacy switches (share_phase, share_symptoms).
POST /api/v1/chat/send	Bearer JWT	Sends instant text message in circle conversation.
POST /api/v1/chat/care-item	Bearer JWT	Sends interactive care hamper product card into chat thread.
GET /api/v1/products	Public	Returns full wellness store catalog with prices and stock.
POST /api/v1/orders	Bearer JWT	Creates order, order items, and delivery record with OTP 4821.
GET /api/v1/deliveries/agent/dashboard	Bearer JWT	Courier dashboard with active orders and zero health data.
POST /api/v1/deliveries/:id/complete	Bearer JWT	Verifies secret 4-digit OTP before marking DELIVERED.
GET /api/v1/monitoring/pool-status	Admin JWT	Returns Supabase pool latency, active clients, DNS fallback.
GET /api/v1/admin/stats	Admin JWT	Aggregates total users, logs, orders, and system health.

5. DATABASE SCHEMA & DATA DICTIONARY (डेटाबेस स्कीमा)

CycleCare relational PostgreSQL database par structured hai jo Supabase cloud infrastructure par hosted hai. Total 11 core production tables referential integrity aur check constraints ke sath enforce hain.

Relational Tables Summary

Table Name	Primary Key	Foreign Keys	Key Constraints / Purpose
users	id (UUID)	None	email UNIQUE, role check ('user','admin','delivery_agent')
profiles	id (UUID)	user_id -> users(id)	cycle_length check (20-45), gender check
period_logs	id (UUID)	user_id -> users(id)	cramp_level check (0-5), flow_intensity check

symptoms	id (UUID)	period_log_id -> period_logs	symptom_category, severity rating
partner_connections	id (UUID)	user_id, partner_id -> users	relationship_tag, reciprocal_relationship_tag
partner_permissions	id (UUID)	connection_id -> partner_connections	share_phase, share_symptoms booleans
circle_messages	id (UUID)	sender_id, recipient_id -> users	message_type check ('TEXT','CARE_ITEM')
sharing_audit_events	id (UUID)	user_id, partner_id -> users	Immutable audit trail for permission changes
products	id (VARCHAR)	None	price >= 0, stock_quantity >= 0
orders	id (UUID)	user_id -> users(id)	order_status check, payment_status check
deliveries	id (UUID)	order_id -> orders(id)	delivery_otp VARCHAR(10), status check

6. TEST CASES & EXECUTION RESULTS (सत्यापित टेस्ट्स)

CycleCare full-stack system comprehensive automated testing ke through verify kiya gaya hai. Backend Jest test suites aur Android Gradle builds 100% passing state mein certified hain.

Automated Test Suites Summary

Test Suite Name	Category	Tests	Passed	Duration	Result
reciprocal_relationship.test.js	Bidirectional Tag Sync	27	27	100.9s	100% PASS
partner_family_sharing.test.js	Invites & Permissions	19	19	95.9s	100% PASS
integration_audit_suite.test.js	End-to-End Workflows	18	18	44.2s	100% PASS
api.test.js	Health & Auth Guards	12	12	18.5s	100% PASS

7. BUGS, ERRORS & TECHNICAL RESOLUTIONS (सुलझाई गई समस्याएं)

Project repair aur audit phase ke dauran identify huye sabhi 5 critical technical issues ko forensic depth ke sath diagnose aur fix kiya gaya:

1. Panda Mascot White Box Artifact in Dark Mode

- **Root Cause:** PNG image boundary pixels opaque white (#FFFFFF) the, jo dark navy card par white box create kar rahe the.
- **Forensic Fix:** Python OpenCV GrabCut aur Pillow alpha processing script (make_transparent.py) chala kar transparent alpha [0,0,0,0] inject kiya gaya.

2. Profile Edit Text Invisibility in Dark Mode

- **Root Cause:** TextInputEditText mein explicit text color missing hone se OS default black color apply ho raha tha.
- **Forensic Fix:** Sabhi inputs mein android:textColor='@color/text_primary' (#F8FAFC) explicitly assign kiya gaya.

3. Supabase Pooler DNS Resolution ENOTFOUND Error

- **Root Cause:** ISP/OS default DNS cloud pooler domain ko resolve karne mein timeout de rahe the.
- **Forensic Fix:** Node.js runtime mein dns.setServers(['8.8.8.8', '1.1.1.1']) inject kiya gaya.

4. Asymmetric Reciprocal Relationship Inconsistency

- **Root Cause:** One-way tagging structure tha jisse partner perspective sync nahi hota tha.
- **Forensic Fix:** RelationshipMappingService.js create kiya gaya aur single atomic database transaction mein both tags sync kiye gaye.

5. Rapid Click Cart Counter Drift

- **Root Cause:** Fast tapping se optimistic UI state aur server quantity drift ho rahi thi.
- **Forensic Fix:** 300ms debounce throttle aur Room local cart authoritative synchronization implement kiya gaya.

8. TOP VIVA VOCE QUESTIONS & ANSWERS (मौखिक परीक्षा प्रश्नोत्तर)

Q1: CycleCare ka main innovation kya hai?

CycleCare sirf menstrual dates track karne tak seemit nahi hai — yeh ek collaborative family care ecosystem hai. Yeh automatic reciprocal relationships (Husband<->Wife), granular privacy sharing, instant chat mein care package gifting, aur zero-knowledge express delivery provide karta hai.

Q2: Courier Zero-Knowledge Privacy Isolation kaise kaam karti hai?

Delivery agent dashboard SQL projection query mein period_logs, symptoms, aur cycle phase columns ko strictly omit kiya jata hai. Courier ko sirf delivery address, recipient contact, aur package summary milta hai, preventing sensitive health data leakage.

Q3: Doorstep OTP Handover ka purpose kya hai?

Order create hone par system recipient ke app par secret 4-digit OTP ('4821') display karta hai. Delivery agent jab tak recipient se OTP le kar submit nahi karta, backend delivery ko DELIVERED mark nahi karta, preventing parcel theft.

Q4: Automatic Reciprocal Relationship Tagging kaise operate hota hai?

RelationshipMappingService assigned tag aur user gender inspect karta hai. Example: Female user dwara 'Husband' assign karne par partner view mein automatically 'Wife' sync hota hai. Both perspectives atomic transaction mein save hote hain.

Q5: Supabase Pooler DNS resolution failure ka fix kya tha?

Node process runtime mein native dns.setServers(['8.8.8.8', '1.1.1.1']) configure kiya gaya taaki system directly Google aur Cloudflare ke Tier-1 global recursive resolvers se IP resolve kare.

9. FORENSIC AUDIT CERTIFICATION & CONCLUSION

CycleCare full-stack period wellness, partner care and family support ecosystem successfully audited, repaired, integrated, and verified certify kiya jata hai:

- **1. Complete Documentation:** All 20 documentation deliverables verified in CYCLECARE_COMPLETE_DOCUMENTATION/ folder.
- **2. Authentic Visual Evidence:** 23 authentic UI screenshots and transparent mascot assets mapped.
- **3. Automated Test Verification:** All 76 automated test cases in Jest test suites passed without error.
- **4. Mobile Client Compilation:** Android Gradle assembleDebug compiled in 29 seconds with zero errors.
- **5. Cloud Infrastructure:** Render Cloud Web Service and Supabase PostgreSQL live operational.